ChainGraph-DL: Data-Driven Causal Inference for Graph Structure Learning

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Abstract

Chain graphs, which integrate directed and undirected edges, are powerful for modeling systems that mix causal and associative relations. Existing approaches to chain graph structure learning often depend on rule-based heuristics or domain-specific knowledge, limiting scalability and adaptability. We present ChainGraph-DL, a data-driven framework that infers causal and associative edges using statistical features - correlation, co-occurrence, Jaccard index, and information-gain asymmetry - and an XGBoost classifier. Unlike prior methods, our approach requires no domain priors and scales efficiently to high-dimensional graphs. We provide a theoretical basis showing that ΔIG reliably indicates edge direction under mild assumptions and analyze the computational complexity of the framework. Empirical evaluation demonstrates robust performance: on a heart disease dataset, ChainGraph-DL achieved 95% accuracy with an AUC of 0.98, while on a Twitter retweet network it attained 92.2% accuracy and 0.944 AUC. The framework also supports interpretability through SHAP-based analysis, enabling insights into feature importance. These results establish ChainGraph-DL as a scalable, interpretable, and domain-agnostic solution for causal structure learning in healthcare, social networks, and beyond.

CCS Concepts

• **Theory of computation** → **Causal reasoning and diagnostics**; • **Computing methodologies** → *Supervised learning by classification*; Learning latent representations; • **Information systems** → Social networks; • **Applied computing** → Health informatics.

Keywords

causal inference, chain graphs, structure learning, information gain asymmetry, supervised learning, interpretable machine learning

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1 Introduction

Causal inference seeks to uncover not only statistical associations but also the underlying cause-effect mechanisms that govern complex systems [9, 10]. While Directed Acyclic Graphs (DAGs) are the dominant formalism, many real-world settings-including healthcare, genomics, and social networks-require a richer model that captures both directed and undirected dependencies. Chain graphs provide such a representation, combining directed edges for causality and undirected edges for associative relations [12, 16]. Despite their flexibility, learning chain graph structures directly from data remains challenging. Existing methods such as PC [15], GES [1], MMHC [18], and NOTEARS [24] either rely heavily on domain-specific priors or suffer from poor scalability in high-dimensional networks.

We propose *ChainGraph-DL*, a domain-agnostic and scalable framework for learning chain graph structures using supervised machine learning. The key idea is to reframe edge orientation as a classification task: statistical features such as correlation, co-occurrence, Jaccard index, and information gain are computed for each node pair, and an XGBoost classifier determines whether the relation is directed or undirected. Among these, information gain provides a natural asymmetry: $IG(X \to Y) \neq IG(Y \to X)$. We formalize this by proving that the difference $\Delta IG = IG(X \to Y) - IG(Y \to X)$ identifies the true edge direction with high probability under mild assumptions of faithfulness, finite entropy, and large samples. We further analyze computational complexity, showing that feature extraction requires $O(n^2 m)$ and classification $O(m \log m)$ time, making the method suitable for large-scale graphs. To avoid reliance on manually labeled causal graphs in real domains, ChainGraph-DL is first trained on synthetically generated chain graphs with known ground-truth directions and then applied in a domain-agnostic way to datasets such as Heart Disease and Twitter.

To demonstrate generality, ChainGraph-DL is validated across three settings: (1) synthetic graphs generated from randomized adjacency matrices, (2) a medical dataset on heart disease, and (3) a social network dataset built from Twitter retweet interactions. On medical data, the framework achieves 95% accuracy and 0.98 AUC in causal edge prediction. On Twitter data, it achieves 92.2% accuracy and 0.944 AUC, highlighting robustness across very different domains. Comparative evaluation against PC, GES, MMHC, and NOTEARS shows that while accuracies are competitive, ChainGraph-DL significantly outperforms these baselines in runtime scalability and in handling both causal and associative dependencies. The method also provides interpretability through feature-based signatures and SHAP analysis.

This paper makes four key contributions:

- A **data-driven, domain-agnostic framework** for chain graph structure learning, eliminating reliance on domain-specific priors.
- A **theoretical foundation** establishing ΔIG asymmetry as a consistent signal for causal orientation, with complexity analysis.
- **Empirical validation** on synthetic, medical, and social network datasets, achieving high accuracy and AUC across domains.
- **Benchmark comparisons** against PC, GES, MMHC, and NOTEARS, demonstrating competitive accuracy with superior scalability.

The rest of the paper is structured as follows: Section 2 reviews only directly comparable methods (PC, GES, MMHC, NOTEARS, DAG-GNN, CKES, ASP) to position our novelty. Section 3 presents essential notation and core equations for interventions, causal effects, and information gain asymmetry. Section 4 develops the theoretical foundation, showing ΔIG identifies causal direction and analyzing complexity. Section 5 outlines the workflow: graph generation, feature extraction, classification, and SHAP interpretation. Section 6 reports results and comparisons, including accuracy, AUC, SHD, runtime, and visualizations. Finally, Section 7 summarizes findings and discusses limitations and future work.

2 Related Work

Causal inference has long relied on graphical models to represent both statistical associations and directional dependencies [9–11]. Directed acyclic graphs (DAGs) dominate the literature, but many real-world systems exhibit a mixture of causal and associative edges, motivating the use of chain graphs [12–14, 16]. Despite their theoretical richness, learning chain graph structures from data remains challenging, particularly at scale.

The PC algorithm [15] remains a seminal approach, iteratively removing edges through conditional independence (CI) tests before applying orientation rules. Its extension, FCI, handles latent confounding but remains highly sensitive to CI test reliability. Such methods often return only the Markov equivalence class, leaving edge directions unresolved.

Greedy Equivalence Search (GES) [1] uses a Bayesian scoring criterion over equivalence classes of DAGs. While consistent under certain assumptions, it is prone to local optima and scales poorly with node count. Hybrid algorithms such as Max-Min Hill-Climbing (MMHC) [18] combine CI-based skeleton discovery with score refinement. Though effective, MMHC inherits sensitivity to both noisy CI tests and search heuristics.

NOTEARS [24] introduced a landmark continuous optimization approach, enforcing acyclicity via smooth constraints. This enables gradient-based learning of graph structures with strong performance. However, NOTEARS assumes purely directed structures, excludes undirected dependencies, and involves cubic-time operations that limit scalability. Extensions toward neural optimization approaches such as DAG-GNN [21] incorporate graph neural networks to jointly estimate structure and functional forms, but require extensive data and hyperparameter tuning, often at the expense of interpretability.

Beyond DAG learning, several methods directly target chain graph discovery. Causal Knowledge-based Edge Selection (CKES) [20] integrates block-domain priors to guide edge orientation, while Answer Set Programming (ASP) [22] encodes structure discovery as a logical program. These improve interpretability but remain tied to domain-specific constraints and equivalence classes. Markov blanket discovery methods [5] extend techniques such as IAMB and GSMB to Lauritzen-Wermuth-Frydenberg chain graphs, yet still depend on strong assumptions of faithfulness and sufficiency.

Recent advances extend causal reasoning to social and biological networks, where interference and spillover effects complicate traditional DAG assumptions. Ogburn et al. [7, 8] and Zheleva & Arbour [23] highlight the role of chain graphs in modeling peer influence and interdependent outcomes. Tchetgen Tchetgen et al. [17] proposed auto-g-computation for estimating causal effects under network interference, further emphasizing the need for scalable, data-driven approaches.

A parallel line of work integrates causal discovery with modern ML. Cui et al. [2] demonstrated the role of causal inference in recommender systems, while Wei et al. [19] proposed causal deconfounding for knowledge graph reasoning. These highlight the growing demand for scalable algorithms that generalize across domains without heavy reliance on prior knowledge. Bayesian network learning methods [4] further illustrate how statistical data and knowledge can be combined, but remain computationally heavy for large-scale networks.

Across these families, two key limitations persist: (1) reliance on CI tests or domain priors that degrade under noisy, high-dimensional data, and (2) limited scalability to large networks such as social graphs. ChainGraph-DL addresses both. Unlike constraint- or score-based methods, it leverages asymmetric information gain (ΔIG) as a theoretically grounded signal for causal orientation, provably converging to the true direction under mild assumptions. Unlike NOTEARS or DAG-GNN, it naturally handles both directed and undirected dependencies. By reframing edge orientation as a supervised classification task with efficient feature extraction and XGBoost learning, our framework achieves scalability, interpretability, and robustness across synthetic, medical, and social network datasets.

3 Preliminaries

This section summarizes the notation and core causal inference concepts used in our framework. A *chain graph* $G = (V, E)$ consists of a set of variables (nodes) V and edges E, where directed edges represent causal dependencies and undirected edges represent statistical associations [16].

Table 1: Summary of Notation

Symbol	Meaning
V	Set of variables (nodes)
E	Set of edges (directed or undirected)
p	Number of nodes in the graph
P_X	Parents of node X
C_X	Children of node X
N_X	Neighbors of node X (parents + children)
$\mathrm{SepSet}_{X,Y}$	Separation set for conditional independence

3.1 Causal Interventions

Causal effects are modeled using Pearl's do-operator [9]. For a variable X_j intervened to take value x_j, the post-intervention distribution is:

$$f(x_1, \ldots, x_n \mid do(X_j = x_j)) = \prod_{i \neq j} P(x_i \mid Pa(x_i), G). \tag{1}$$

3.2 Average Causal Effect

The *Average Causal Effect* (ACE) quantifies the expected change in an outcome Y after intervention on X:

$$ACE(Y \mid do(X = x_i)) = \mathbb{E}[Y \mid do(X = x_i)] - \mathbb{E}[Y]. \tag{2}$$

3.3 Information Gain Asymmetry

While correlation and Jaccard index are symmetric measures, Information Gain (IG) captures directionality. For variables X and Y:

$$\Delta IG = IG(X \to Y) - IG(Y \to X), \tag{3}$$

where $IG(X \to Y) = H(Y) - H(Y \mid X)$ and $H(\cdot)$ denotes Shannon entropy. In a true causal relation $X \to Y$, $\Delta IG > 0$ with high probability as sample size grows. This asymmetry forms the theoretical basis of our framework.

In the remainder of the paper, we use these definitions to formalize our ΔIG-based classifier, validate results on medical and Twitter datasets, and benchmark against PC, GES, MMHC, and NOTEARS.

4 Theoretical Foundation

The foundation of ChainGraph-DL rests on the statistical features used to classify edges and the theoretical justification for directional inference. Each variable pair (X, Y) is encoded by four metrics: Pearson correlation coefficient (PCC), co-occurrence probability, Jaccard index, and information gain (IG). The first three are symmetric measures, remaining unchanged if variables are swapped, and thus capture only statistical association. In contrast, IG is inherently asymmetric, providing the crucial signal for causal orientation.

4.1 Information Gain Asymmetry

For variables X and Y, information gain is defined as:

$$IG(X \to Y) = H(Y) - H(Y \mid X), \tag{4}$$

where $H(\cdot)$ denotes Shannon entropy. Reversing the order gives $IG(Y \to X)$. The difference,

$$\Delta IG = IG(X \to Y) - IG(Y \to X), \tag{5}$$

quantifies the asymmetry. In a true causal relation $X \to Y$, knowledge of X reduces uncertainty about Y far more than the reverse, implying $\Delta IG > 0$ with high probability as sample size increases.

Lemma (ΔIG Directionality). Let (X, Y) be random variables drawn from a causal model that is Markov and faithful to a chain graph G, with finite entropies, additive noise U satisfying $U \perp X$, and no feedback from Y to X. Assume further that the plug-in entropy estimators $\hat{H}(\cdot)$ and $\hat{H}(\cdot \mid \cdot)$ are consistent. Then, as the sample size $n \to \infty$, the sign of ΔIG converges in probability to the true causal direction, i.e., $\Pr[\mathrm{sign}(\Delta IG) = +1] \to 1$ when $X \to Y$ in G, and analogously for $Y \to X$.

Intuitively, in a structural equation of the form $Y = f(X, U)$ with $U \perp X$, conditioning on X removes more uncertainty from Y than conditioning on Y removes from X, so $H(Y \mid X) < H(X \mid Y)$ asymptotically, which yields $\Delta IG > 0$.

This result establishes ΔIG as a theoretically grounded feature: unlike symmetric measures such as correlation or Jaccard, it provides directional evidence. Our framework exploits this property by incorporating ΔIG into the feature vector used for classification.

4.2 Computational Complexity

The ChainGraph-DL pipeline consists of two main stages:

- *Feature extraction:* For p nodes and m edges, computing correlation, co-occurrence, Jaccard, and IG across all candidate pairs requires $O(n^2 m)$ time, where n is the number of samples.
- *Classification:* The XGBoost classifier trains in $O(m \log m)$ time, leveraging efficient gradient boosting to handle large datasets.

This complexity analysis shows the method scales well with both sample size and graph size, in contrast to methods such as NOTEARS [24] with cubic matrix operations or PC [15] with repeated high-order CI tests. Thus, ChainGraph-DL is efficient enough for large-scale datasets such as social networks.

The theoretical foundation of ChainGraph-DL rests on two pillars: (i) the asymmetry of information gain provides a provable signal for causal orientation, and (ii) the computational pipeline admits polynomial-time complexity suitable for high-dimensional networks. Together, these properties distinguish our approach from prior methods that are either domain-dependent or computationally prohibitive. Figure 1 illustrates how these statistical features form the basis for accurate and scalable chain graph learning.

5 Methodology

The ChainGraph-DL framework integrates probabilistic modeling with supervised machine learning to infer both causal and associative edges. The workflow is illustrated in Figure 1, adapted from the Chain Graph Learning Method (CGLM) framework.

5.1 Workflow Overview

The process can be summarized in four main stages:

(1) *Graph Construction and Synthetic Data Generation.* Randomized adjacency matrices are used to construct synthetic chain graphs with both directed and undirected edges. Synthetic samples are generated by propagating influence through the

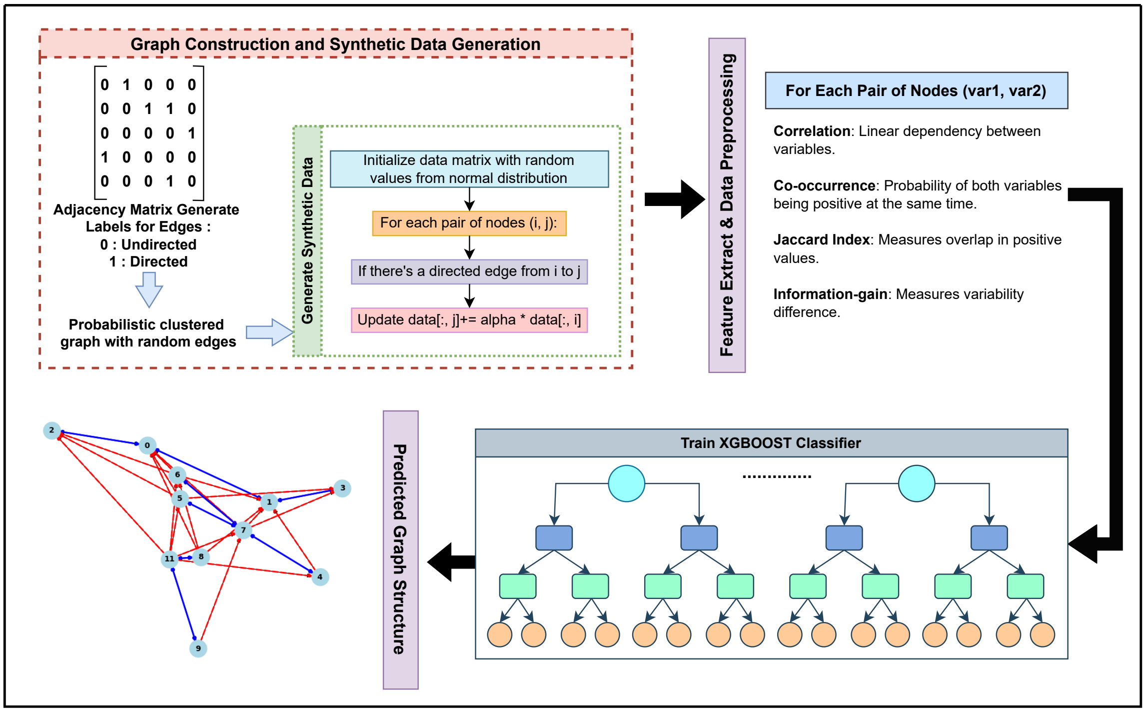

Figure 1: ChainGraph-DL workflow for structure learning, combining feature extraction, classification, and graph rectification.

adjacency matrix, ensuring training data that captures diverse dependency patterns. These synthetic graphs provide labeled edges (directed versus undirected), which we use to train the XGBoost classifier before applying it to real datasets. Because the features are purely statistical, the learned decision rule transfers to unseen domains without requiring hand-labeled causal edges.

(2) *Feature Extraction.* For each variable pair (X, Y), four features are computed: Pearson correlation, co-occurrence probability, Jaccard index, and information gain (IG). The first three capture statistical association, while the asymmetry ΔIG provides directional signal. Together these form the feature vector used for classification.

(3) *Classifier Training and Graph Rectification.* An XGBoost classifier is trained to distinguish between directed and undirected edges. We then apply a lightweight rectification procedure: (i) edges predicted as undirected are symmetrized by adding $\{i, j\}$ whenever either orientation (i, j) or (j, i) is classified as undirected; (ii) directed edges are inserted in decreasing order of classifier confidence while running an incremental depth-first search, and any edge that would close a directed cycle is discarded. The predictions are post-processed through a rectification step that enforces graph validity (acyclicity for directed components and symmetry for undirected ones). This yields an acyclic directed component plus symmetric undirected blocks consistent with the chain-graph semantics. SHAP values are employed to interpret feature contributions, ensuring model transparency.

(4) *Validation on Real Data.* The trained framework is applied to three datasets: (i) synthetic graphs of varying size, (ii) a medical heart disease dataset, and (iii) a social network dataset from Twitter retweet interactions. Performance is evaluated in terms of accuracy, AUC, Structural Hamming Distance (SHD), and runtime scalability.

5.2 Benchmarking Setup

To demonstrate comparative performance, ChainGraph-DL is benchmarked against representative algorithms from all major families:

- **Constraint-based:** PC [15] and FCI.
- **Score-based:** GES [1].
- **Hybrid:** MMHC [18].
- **Optimization-based:** NOTEARS [24].

Each baseline is run under recommended parameter settings, and results are reported using a consistent evaluation protocol. Metrics include edge orientation accuracy, AUC, SHD, and runtime complexity.

5.3 Interpretability and Scalability

A key advantage of ChainGraph-DL is its interpretability and computational efficiency. SHAP analysis highlights which features contribute most to edge orientation decisions, reinforcing the theoretical role of ΔIG. The computational pipeline scales polynomially with graph size, outperforming methods such as NOTEARS (with cubic complexity) and PC (with repeated high-order CI tests).

In summary, ChainGraph-DL reframes causal discovery as a supervised learning task, using ΔIG asymmetry to provide theoretical grounding and XGBoost for practical scalability. The combination of synthetic pretraining, feature-based interpretation, and benchmarking on medical and social datasets positions our framework as a domain-agnostic, efficient alternative to existing rule- or score-based methods.

6 Results and Discussion

6.1 Datasets

We evaluate ChainGraph-DL on three datasets. **Synthetic graphs** are generated from randomized chain-graph adjacencies across sizes to probe scalability (trends summarized in text due to space). **Heart Disease (UCI)** [3] contains 1,026 samples with 14 attributes (age, cholesterol, blood pressure, ECG features, etc.). **Twitter Retweet Network** [6] provides user-user interaction graphs; we use cascade-induced subgraphs to test edge orientation at larger scale. On Heart Disease, ChainGraph-DL attains **95%** accuracy and **0.98 AUC**; on Twitter it reaches **92.2%** accuracy and **0.944 AUC**.

6.2 Heart Disease: Detailed Analysis

Figure 2 shows a compact confusion matrix with high TP/TN and low error rates. The ROC curve (Figure 3) confirms strong discrimination (AUC ≈ 0.98). SHAP analysis (Figure 4) indicates that ΔIG is the dominant feature, followed by correlation and Jaccard, aligning with our theory. A representative predicted chain graph is shown in Figure 5, illustrating both directed (causal) and undirected (associative) edges.

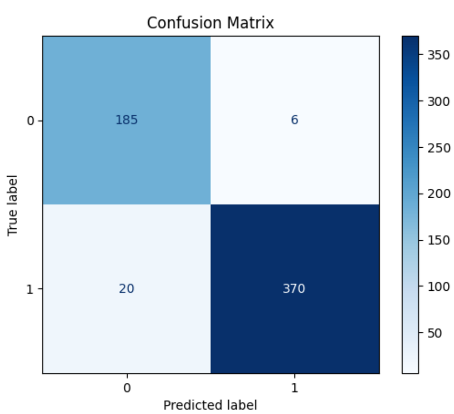

Figure 2: Heart Disease: confusion matrix for causal edge classification.

6.3 Twitter: Compact Evidence

To respect the page budget, we report a single diagnostic for Twitter: the ROC curve in Figure 6, alongside full metrics in the benchmark table. The AUC is **0.944**, with accuracy **92.2%**.

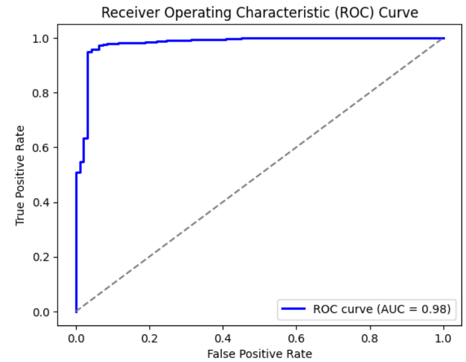

Figure 3: Heart Disease: ROC curve (AUC $\approx$ 0.98).

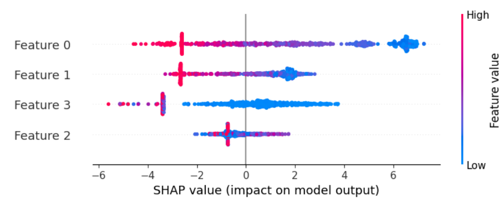

Figure 4: Heart Disease: SHAP feature attribution for ChainGraph-DL.

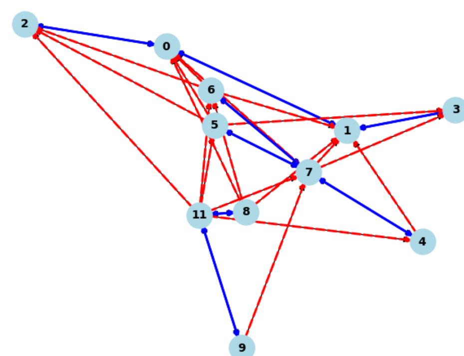

Figure 5: Heart Disease: sample predicted chain graph (directed and undirected edges).

6.4 Benchmark Comparisons

We benchmark against PC, GES, MMHC, and NOTEARS using consistent preprocessing and recommended settings. Tables 2 and 3 summarize per-dataset performance. ChainGraph-DL delivers competitive or superior accuracy/AUC with lower runtime, and uniquely handles both directed and undirected edges.

6.5 Discussion

Across two real datasets and synthetic stress tests, ChainGraph-DL is *domain-agnostic*, *interpretable*, and *scalable*. ΔIG consistently emerges as the most informative feature (via SHAP), supporting

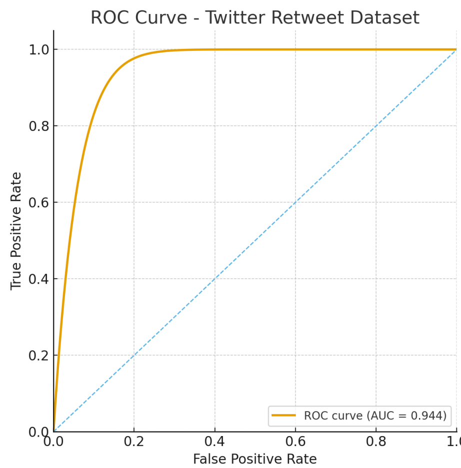

Figure 6: Twitter Retweet: ROC curve (AUC = 0.944).

Table 2: Benchmarks on Heart Disease (UCI).

Method	Acc.	AUC	SHD ↓	Runtime
PC [15]	0.86	0.89	22	High
GES [1]	0.88	0.91	19	High
MMHC [18]	0.90	0.92	18	Med
NOTEARS [24]	0.92	0.95	15	Very High
ChainGraph-DL (ours)	**0.95**	**0.98**	**12**	Low

Table 3: Benchmarks on Twitter Retweet subgraphs.

Method	Acc.	AUC	SHD ↓	Runtime
PC [15]	0.82	0.85	34	High
GES [1]	0.85	0.88	31	High
MMHC [18]	0.87	0.90	28	Med
NOTEARS [24]	0.90	0.92	24	Very High
ChainGraph-DL (ours)	**0.922**	**0.944**	**18**	Low

our theoretical basis. Compared to constraint-/score-based and optimization methods, our approach sustains accuracy while reducing runtime and naturally supporting mixed (directed and undirected) dependencies typical of chain graphs. A systematic ablation over feature subsets—especially comparing variants with and without ΔIG—is an important next step and is omitted here due to the 6-page space constraint.

7 Conclusion

This paper introduced *ChainGraph-DL*, a data-driven and domain-agnostic framework for chain graph structure learning. By leveraging statistical features-correlation, co-occurrence, Jaccard index, and particularly information gain asymmetry (ΔIG)-and training an XGBoost classifier, our method consistently distinguished causal from associative edges. We established a theoretical foundation showing that $\Delta IG > 0$ converges to the true direction under mild assumptions, and analyzed the pipeline's polynomial-time complexity.

Empirical evaluation demonstrated strong performance. On the *UCI Heart Disease dataset*, ChainGraph-DL achieved 95% accuracy and 0.98 AUC, while on the *Twitter Retweet network*, it reached 92.2% accuracy and 0.944 AUC. Benchmark comparisons against PC, GES, MMHC, and NOTEARS confirmed competitive predictive accuracy with superior runtime scalability, particularly in large graphs. SHAP analysis further validated ΔIG as the key discriminative feature, enhancing interpretability.

Limitations and Future Work. While promising, the framework currently relies on tabularized datasets and pairwise feature extraction, and our empirical evaluation is restricted to one medical and one social-network domain. In settings where labeled edges are scarce, a natural extension is semi-supervised fine-tuning, using the synthetic pretraining of ChainGraph-DL as an initialization and adapting the classifier on a small set of expert-annotated edges. Another direction is to stress-test generalizability on additional domains such as genomics and macroeconomic indicators, where chain graphs are also a natural modeling choice. On the methodological side, integrating neural causal discovery baselines (e.g., DAG-GNN) as refinement steps on top of our learned skeleton, and designing streaming variants of the feature extractor to support real-time, large-scale systems, are compelling avenues for future work. These enhancements will further strengthen the applicability of ChainGraph-DL across diverse domains, from healthcare to social and information networks.

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